

Jack <> Bluetooth Design Specification/Details

Purpose

The purpose of the *Jack <> Bluetooth Design Specification* is to document the design choices and provide details on this project.

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1. Jack2Bluetooth features

This project aims to design a device with the following features:

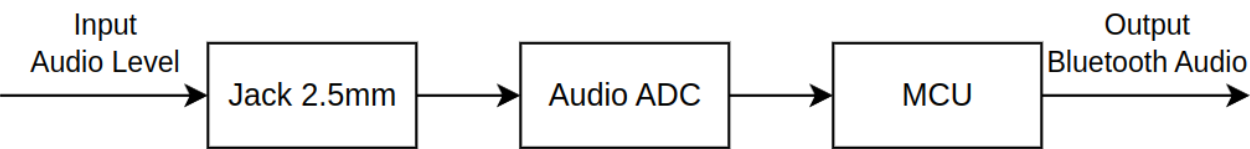
- Audio input from a 2.5mm audio jack can be sampled and transmitted via Bluetooth to a headphone.

- Digital audio from a smartphone can be received via Bluetooth and converted into analog audio through a 3.5mm audio jack.
- The device can be powered via USB-C or run on battery for up to 3 hours.
- The battery charges when the device is connected to USB-C.
- A screen and buttons allow switching between "Jack to Bluetooth" and "Bluetooth to Jack" modes.
- The screen and buttons enable Bluetooth peripheral selection when the device functions as a Bluetooth central.
- The screen turns off after 1 minute of inactivity to save power.

2. Architecture

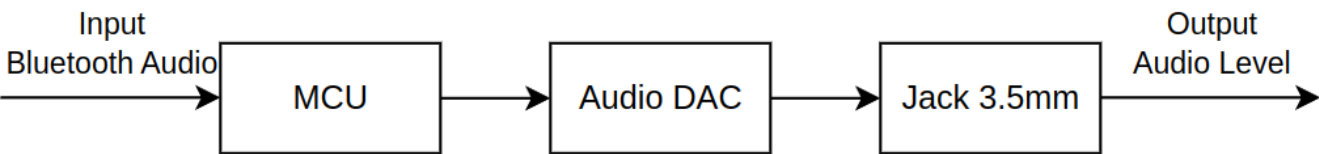
2.1 Jack to Bluetooth

For audio sampling:



The audio codec must support line-level audio input signals. The ADC is integrated into the audio codec with an amplifier.

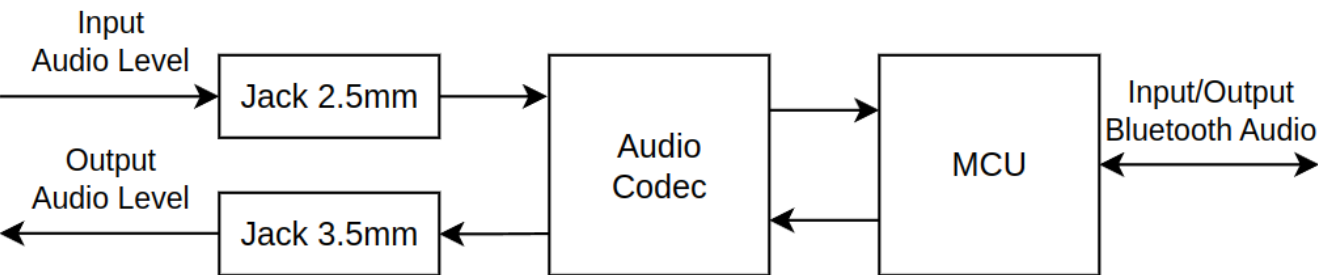
2.2 Bluetooth to Jack



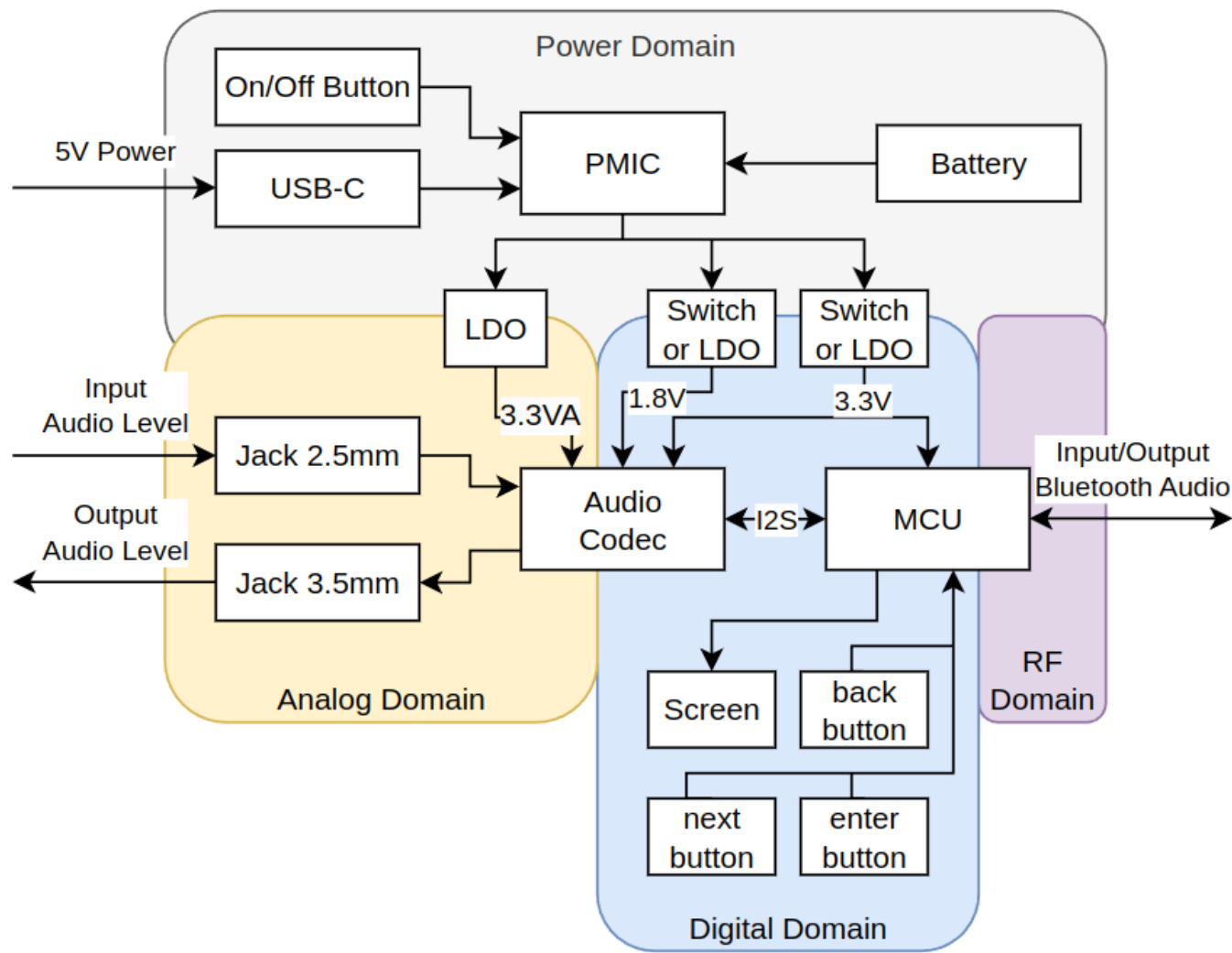
The audio codec must include a DAC with an amplifier.

2.3 Overall architecture

When merging the two previous architecture, it becomes:



And when adding power components and the user interface components, the overall architecture is :



2.4 User Interface

Based on the architecture, the user interface is then:

User interface	Direction
Button Next	Input
Button Enter	Input
Button Back	Input
Button On/Off	Input
Audio Jack 2.5mm	Input
Usb-C	Input
Bluetooth	Input & Output
Screen	Output
Audio Jack 3.5mm	Output

3. Component choices

3.1 MCU : Microcontroller Unit

The ESP32 was chosen due to its widespread availability and ease of use. To simplify RF design, an ESP32 module with an integrated antenna was selected.

Requirements:

- Bluetooth support
- Integrated antenna
- USB, I2S, I2C, SPI inerfaces
- Core frequency (not the max supported but the integrated clock frequency)
- Memory size

Final choice: **ESP32-C-WROOM-N8**.

3.2 Audio Codec

The audio codec must include a DAC and an ADC, support line-level audio, and have single-ended inputs and outputs.

Final choice : **TLV320AIC3101** (Texas Instrument).

3.3 Battery

The battery should provide at least 3 hours of operation. The most power-intensive components are the MCU, audio codec, and the screen.

The screen is going to be on only when a user push a button (and stays on for ~30s). Therefore, it should be off in idle mode, in transmitting mode or receiving mode. Since the screen is mostly off, its consumption is negligible.

3.3.1 ESP32-C6 Current consumption

According to the esp32-c6 datasheet, the Bluetooth LE current consumption are:

Table 5-8. Current Consumption for Bluetooth LE in Active Mode

Work Mode	RF Condition	Description	Peak (mA)
Active (RF working)	TX	Bluetooth LE @ 20.0 dBm	315
		Bluetooth LE @ 9.0 dBm	190
		Bluetooth LE @ 0 dBm	130
		Bluetooth LE @ -15.0 dBm	94
	RX	Bluetooth LE	71

3.3.2 TLV320AIC3101 Current consumption

According to its datasheet, its current consumptions are :

CURRENT CONSUMPTION – DRVDD = AVDD_DAC = IOVDD = 3.3 V, DVDD = 1.8 V				
I_{IN}	$I_{DRVDD} + I_{AVDD_DAC}$	$\overline{RESET}$ held low	0.1	μA
	I_{DVDD}		0.2	
	$I_{DRVDD} + I_{AVDD_DAC}$	Mono ADC record, $f_S = 8$ ksps, I ² S slave, AGC off, no signal	2.15	mA
	I_{DVDD}		0.48	
	$I_{DRVDD} + I_{AVDD_DAC}$	Stereo ADC record, $f_S = 8$ ksps, I ² S slave, AGC off, no signal	4.1	
	I_{DVDD}		0.62	
	$I_{DRVDD} + I_{AVDD_DAC}$	Stereo ADC record, $f_S = 48$ ksps, I ² S slave, AGC off, no signal	4.31 ⁽⁶⁾	
	I_{DVDD}		2.45 ⁽⁶⁾	
	$I_{DRVDD} + I_{AVDD_DAC}$	Stereo DAC playback to lineout, analog mixer bypassed, $f_S = 48$ ksps, I ² S slave	3.5	
	I_{DVDD}		2.3	
	$I_{DRVDD} + I_{AVDD_DAC}$	Stereo DAC playback to lineout, $f_S = 48$ ksps, I ² S slave, no signal	4.9	
	I_{DVDD}		2.3	
	$I_{DRVDD} + I_{AVDD_DAC}$	Stereo DAC playback to stereo single-ended headphone, $f_S = 48$ ksps, I ² S slave, no signal	6.7	
	I_{DVDD}		2.3	
	$I_{DRVDD} + I_{AVDD_DAC}$	Stereo linein to stereo lineout, no signal	3.11	
	I_{DVDD}		0	
	$I_{DRVDD} + I_{AVDD_DAC}$	Extra power when PLL enabled	1.4	
	I_{DVDD}		0.9	
	$I_{DRVDD} + I_{AVDD_DAC}$	All blocks powered down. Headset detection enabled, headset not inserted	28	μA
	I_{DVDD}		2	

So it is negligible compared to the ESP32-C6 Bluetooth LE transmission currents.

3.3.3 Current consumption conclusions

The critical use case for current consumption is during audio sampling and Bluetooth transmission. Assuming the transmission is done at 9dBm, the required battery to power the device 3 hours is around 600mA.

3.4 PMIC : Power Management Integrated Circuit

The PMIC should support power path between the USB-C and the battery, battery charging when USB-C is plugged, battery discharging when unplugged. Since a power on/off system is needed, if this IC could have sleep or deep sleep mode with very low current consumption, it could be nice.

Final choice : **BQ25619RTWR** (Texas Instrument).

All the elements are providing in the first page of product specification:



BQ25618, BQ25619

SLUSDF3E – JUNE 2019 – REVISED JULY 2024

BQ25618/619 I²C Controlled 1-Cell 1.5A Battery Charger with 20mA Termination and 1A Boost Operation

1 Features

- Single chip solution to charge wearable accessories from adapter or battery
- High-efficiency, 1.5MHz, synchronous switch-mode buck charger
 - 95.5% charge efficiency at 0.5A and 94.5% efficiency at 1A
 - ±0.5% charge voltage regulation (10mV step)
 - I²C programmable JEITA profile of charge voltage, current and temperature thresholds
 - Low termination current with high accuracy 20mA±10mA
 - Small inductor form factor of 2.5 x 2.0 x 1.0mm³
- Boost mode with output from 4.6V to 5.15V
 - 94% boost efficiency at 1A output
 - Integrated control to switch between Charge and Boost mode
 - PMID_GOOD pin control external PMOS FET for protection against fault conditions
- Single input supporting USB input, high-voltage adapter, or wireless power
 - Support 4V to 13.5V input voltage range with 22V absolute max input rating
 - Programmable input current limit (IINDPM) with I²C (100mA to 3.2A, 100mA/step)
 - Maximum power tracking by input voltage limit (VINDPM) up to 5.4V
 - VINDPM threshold automatically tracks battery voltage
- Narrow VDC (NVDC) power path management
 - System instant-on with no battery or deeply discharged battery
- Flexible I²C configuration and autonomous charging for optimal system performance
- High integration includes all MOSFETs, current sensing and loop compensation
- Low R_{DS(ON)} 19.5mΩ BATFET to minimize charging loss and extend battery run time
 - BATFET control for Ship mode, and full system reset with and without adapter
- 7µA low battery leakage current in Ship mode
- 9.5µA low battery leakage current with system standby
- High accuracy battery charging profile
 - ±6% charge current regulation
 - ±7.5% input current regulation
 - Remote battery sensing to charge faster
 - Programmable top-off timer for full battery charging

- Safety Related Certifications:
 - IEC 62368-1 CB Certification

2 Applications

- Consumer wearables, smartwatch
- Personal care and fitness
- Headsets/headphone
- Earbuds (True Wireless or TWS) charging case
- Hearing aids charging case

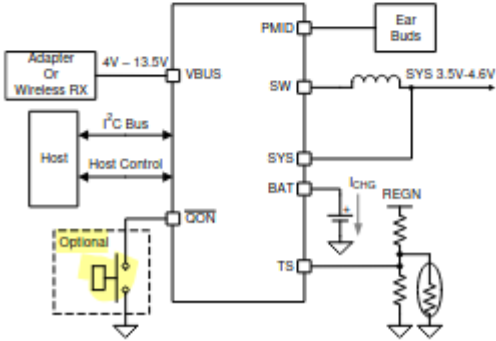
3 Description

The BQ25618/619 integrates charge, boost converter and voltage protection in a single device. It offers the industry's lowest termination current for switching chargers to charge wearable devices by full battery capacity. The BQ25618/619 best-in-class low quiescent current reduces battery leakage down to 6µA in Ship mode, which conserves battery energy to double the shelf life for the device. The BQ25619 is in a 4mm x 4mm QFN package for easy layout. The BQ25618 is in a 2.0mm x 2.4mm WCSP package for space limited design.

Package Information

PART NUMBER	PACKAGE ⁽¹⁾	PACKAGE SIZE ⁽²⁾	BODY SIZE (NOM)
BQ25618	YFF (DSBGA, 30)	2.20mm x 2.60mm	2.00mm x 2.40mm
BQ25619	RTW (WQFN, 24)	4.00mm x 4.00mm	4.00mm x 4.00mm

(1) For all available packages, see Section 13.
(2) The package size (length x width) is a nominal value and includes pins, where applicable.



Simplified Application

3.5 LDO : Low-dropout voltage regulator

It was decided to use an LDO for the analog to have a clean alimentation. LDO should not be fixed to be able to reuse it in other designs.

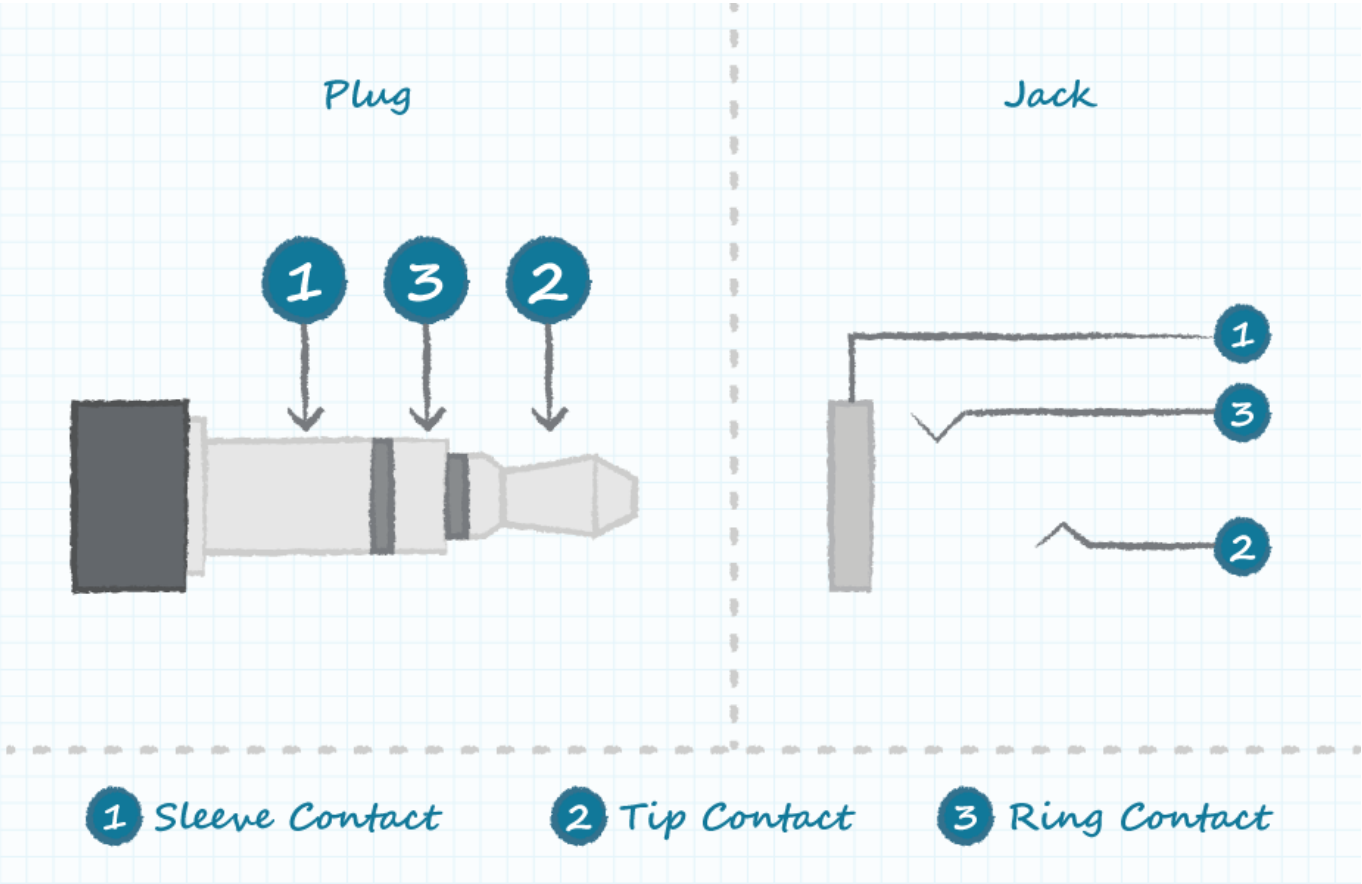
Choice was made to use **TLV758P** from Texas Instrument.

3.6 Switching Voltage Regulator switch

Choice was made to use the **TPS62A01PDDCR** from Texas Instrument.

4. Electronic schematic

4.1 Jack 3.5mm

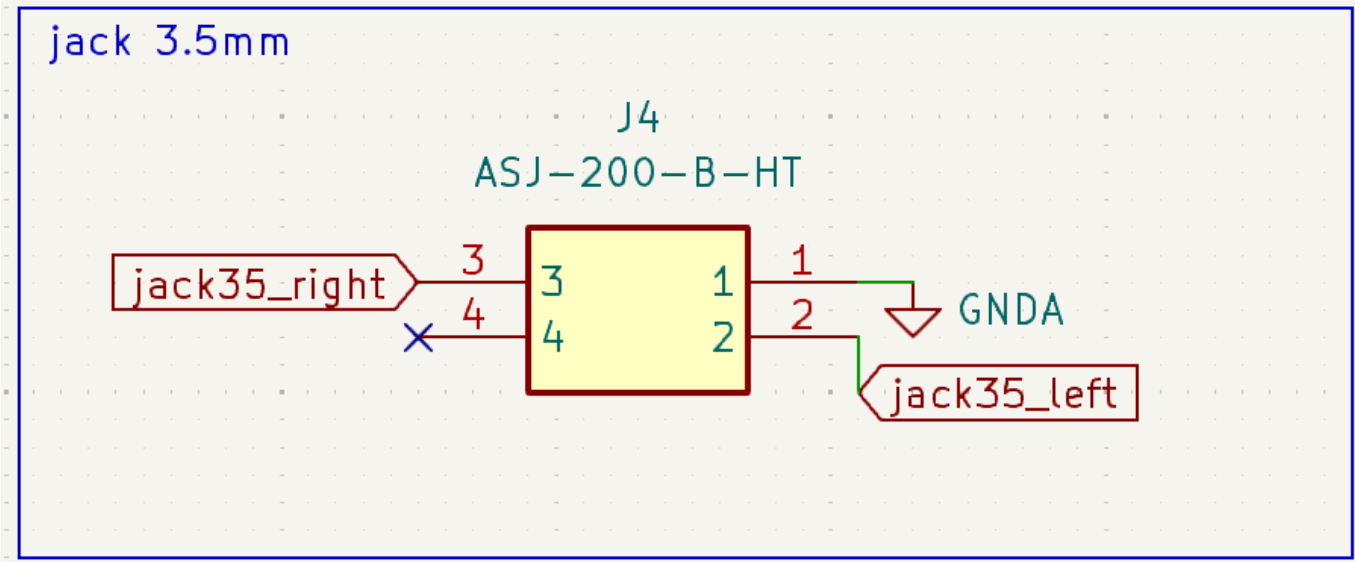


And the ASJ datasheet gives:

Model No.	ASJ-200-B-HT
Schematic	The schematic diagram shows four terminals labeled 1, 3, 4, and 2. Terminal 1 is connected to terminal 3. Terminal 4 is connected to terminal 2. The connections are shown as a continuous line forming a loop.

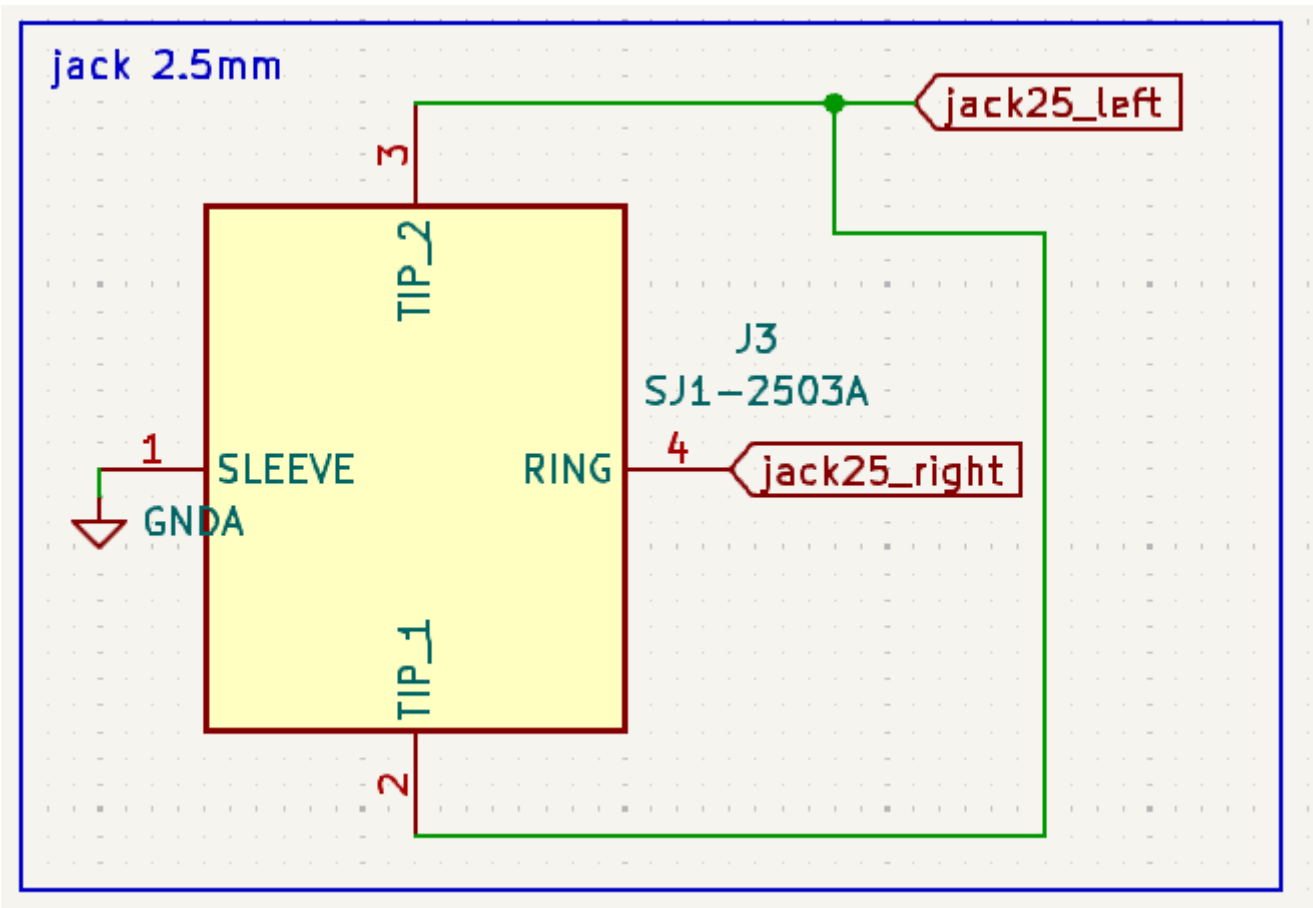
- 1. Sleeve is GND
- 2. Tip is Left
- 3. Ring is Right

So the schematic on kicad is:



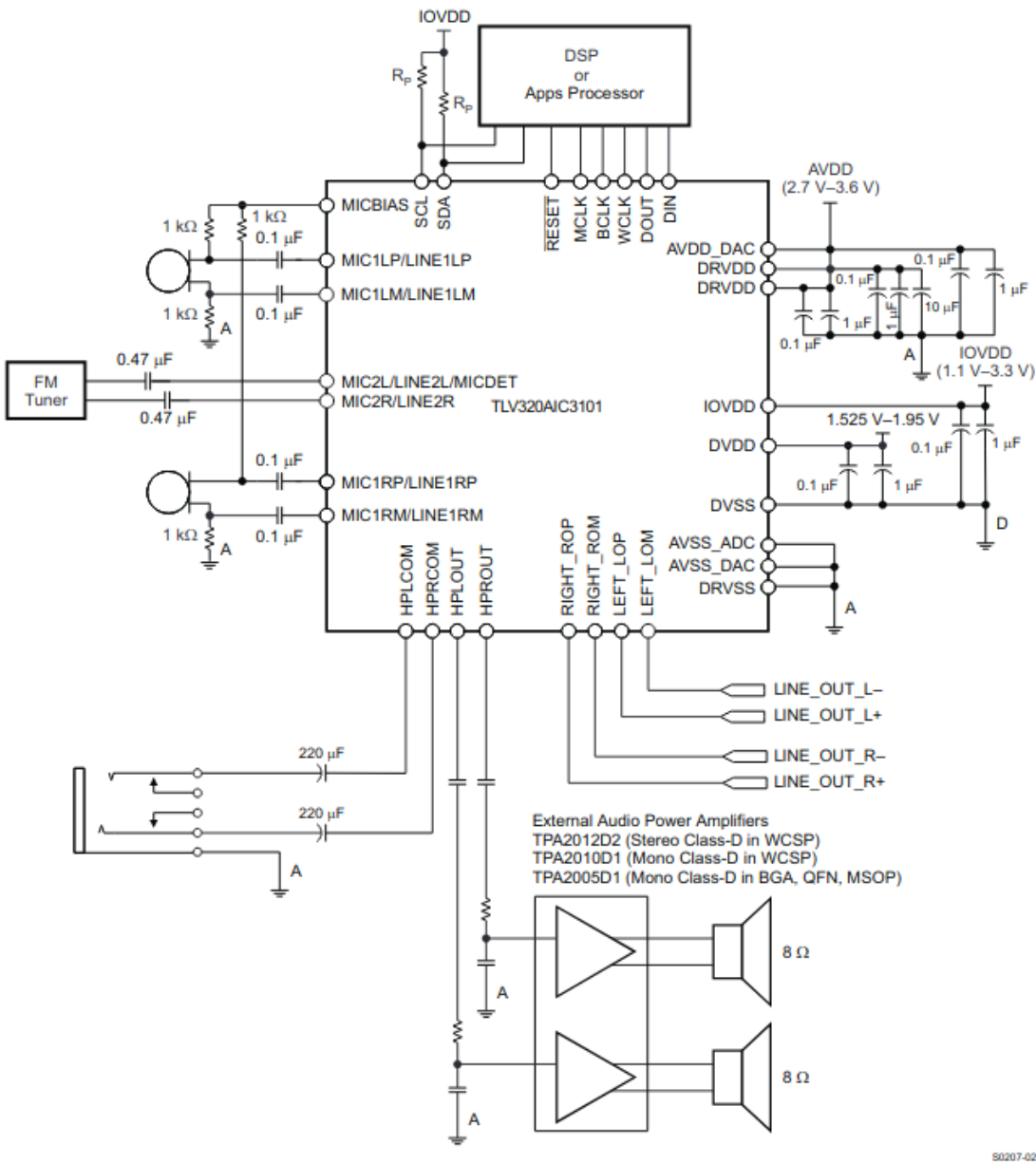
4.2 Jack 2.5mm

The schematic is



4.3 Audio Codec (TLV320AIC3101)

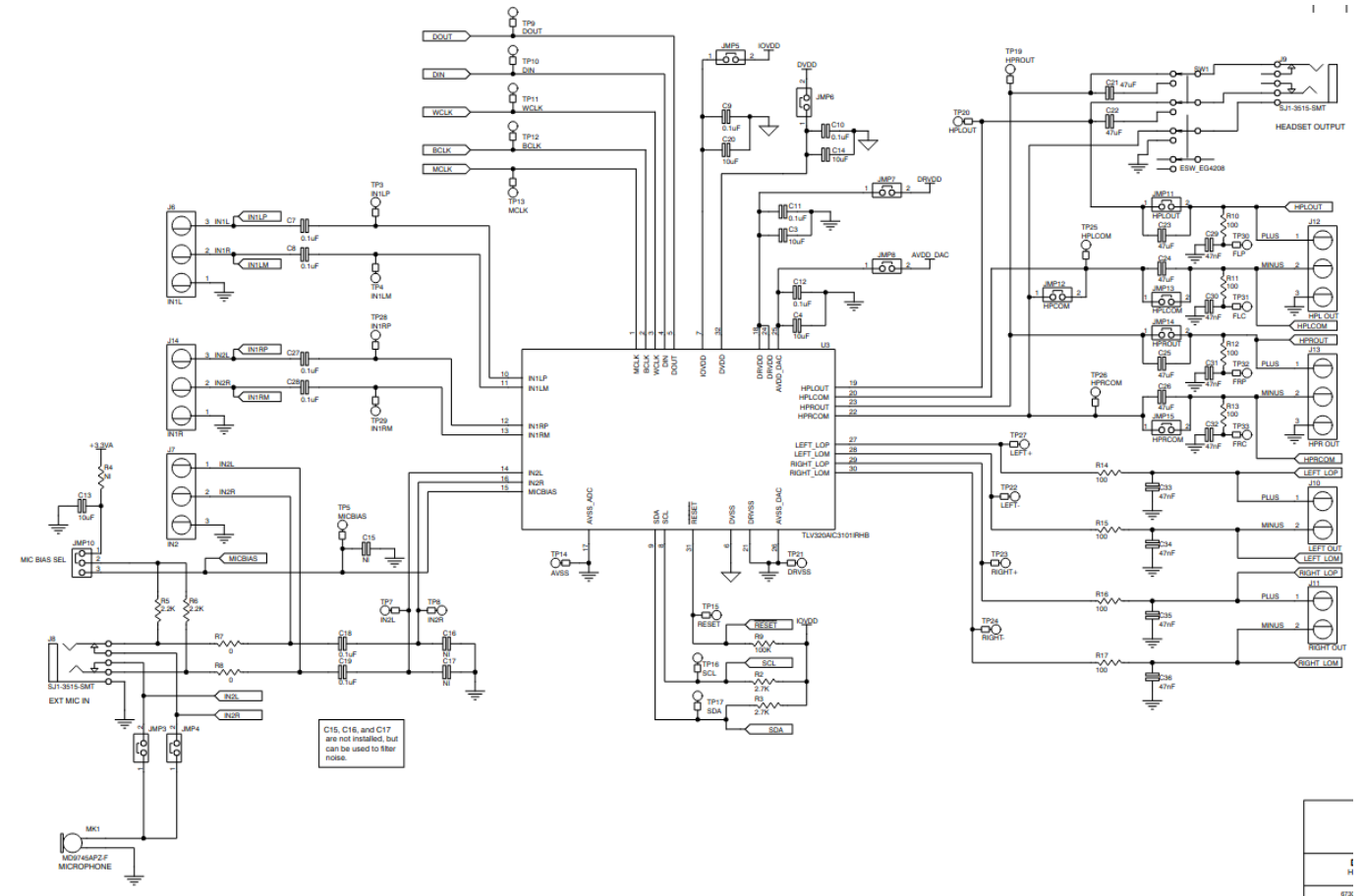
The typical application from the spec is:



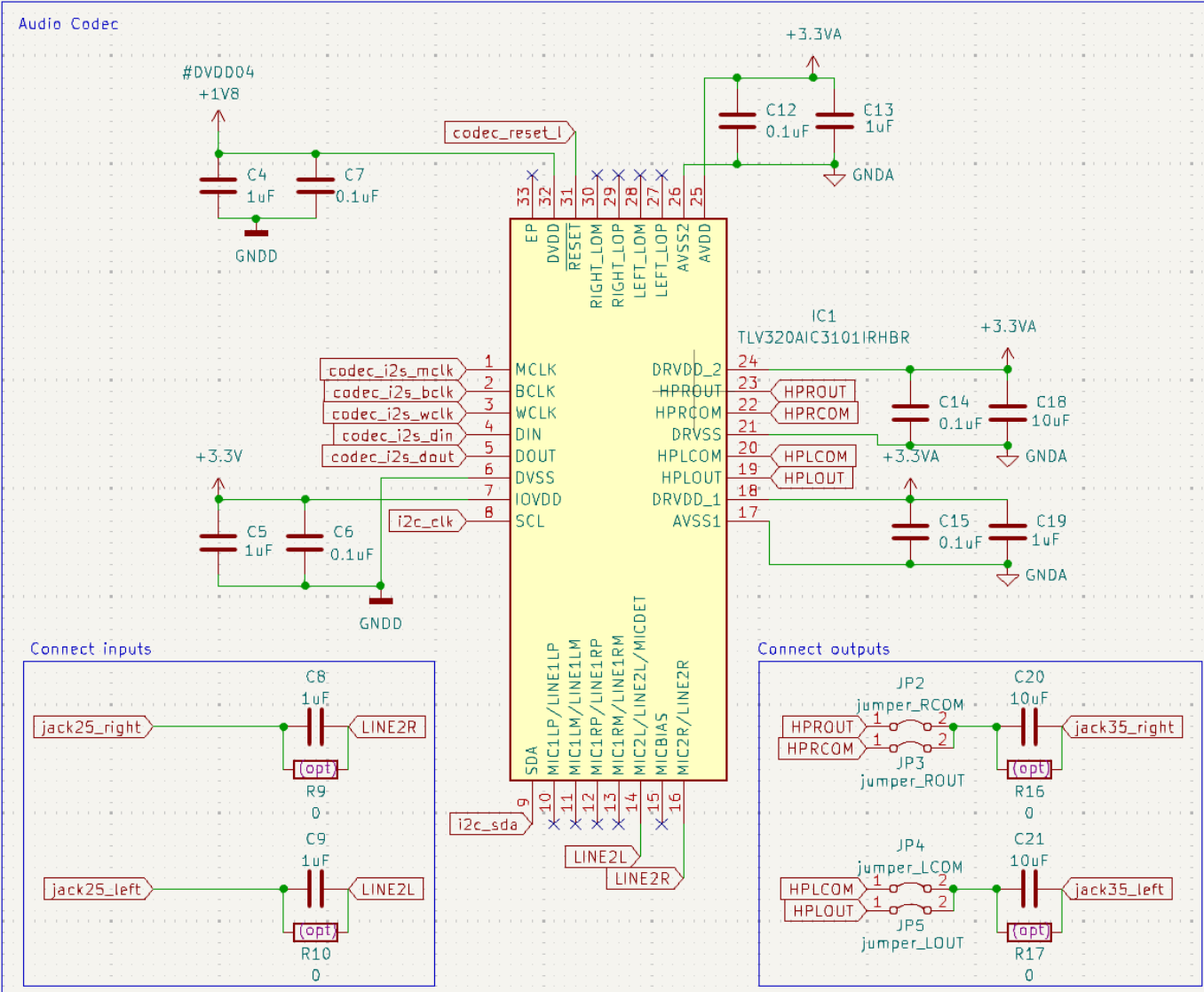
S0207-02

Figure 33. Typical Connections for AC-Coupled Headphone Output With Separate Line Outputs and External Speaker Amplifier

It is also worth looking at the development board:



The schematic done is then:



4 capacitors are to be determined, C8, C9, C20, C21.

First, C8=C9 and C20=C21.

Then, C8 and Rinput_of_line2R is a high pass filter, where $f_c = 1/(2\pi C \cdot R)$.

Input resistance	MIC1L/MIC1R inputs routed to single ADC input mix attenuation = 0 dB	20	kΩ
	MIC1L/MIC1R inputs routed to single ADC input mix attenuation = 12 dB	80	
	MIC2L/MIC2R inputs routed to single ADC input mix attenuation = 0 dB	20	
	MIC2L/MIC2R inputs routed to single ADC input mix attenuation = 12 dB	80	
	MIC3L/MIC3R inputs routed to single ADC input mix attenuation = 0 dB	20	
	MIC3L/MIC3R inputs routed to single ADC input mix attenuation = 12 dB	80	

From TLV320AIC3101 datasheet, Rinput_of_line2R is between 20kΩ and 80kΩ depending on the attenuation programmed in the audio codec.

The audio bandwidth is 20-20kHz. A capacitor of 1uF gives $f_c = 8\text{Hz}$ for $R=20\text{k}\Omega$ and $f_c= 2\text{Hz}$ for $R=80\text{k}\Omega$ which should be good for this application. So C8=C9=1uF.

Under the hypothesis that the load resistance on the jack 3.5mm is $10k\Omega$ (same hypothesis as in the TLV320AIC3101 datasheet), C20 and Rload are doing an high pass filter. A $10\mu F$ capacitor ensures that for $R_{load} > 800\Omega$, the audio bandwidth is respected. So $C20=C21=10\mu F$.

4.4 MCU (ESP32-C6-WROOM-1)

The datasheet of the ESP32-C6-WROOM-1 recommend the following schematic:

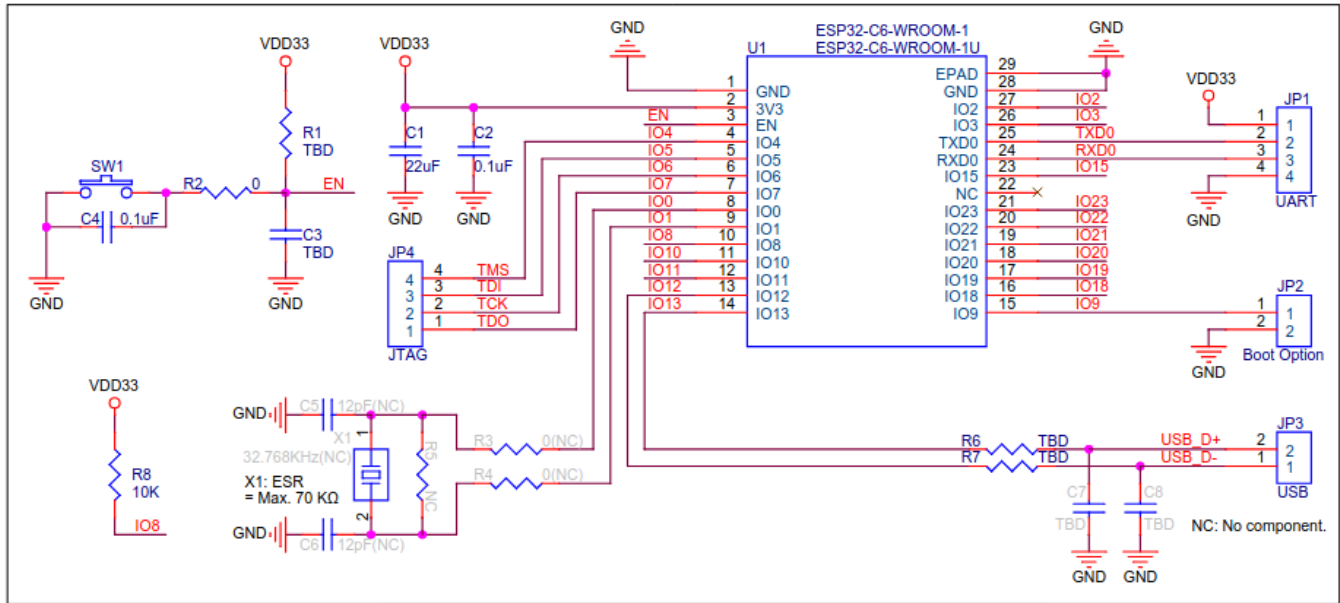
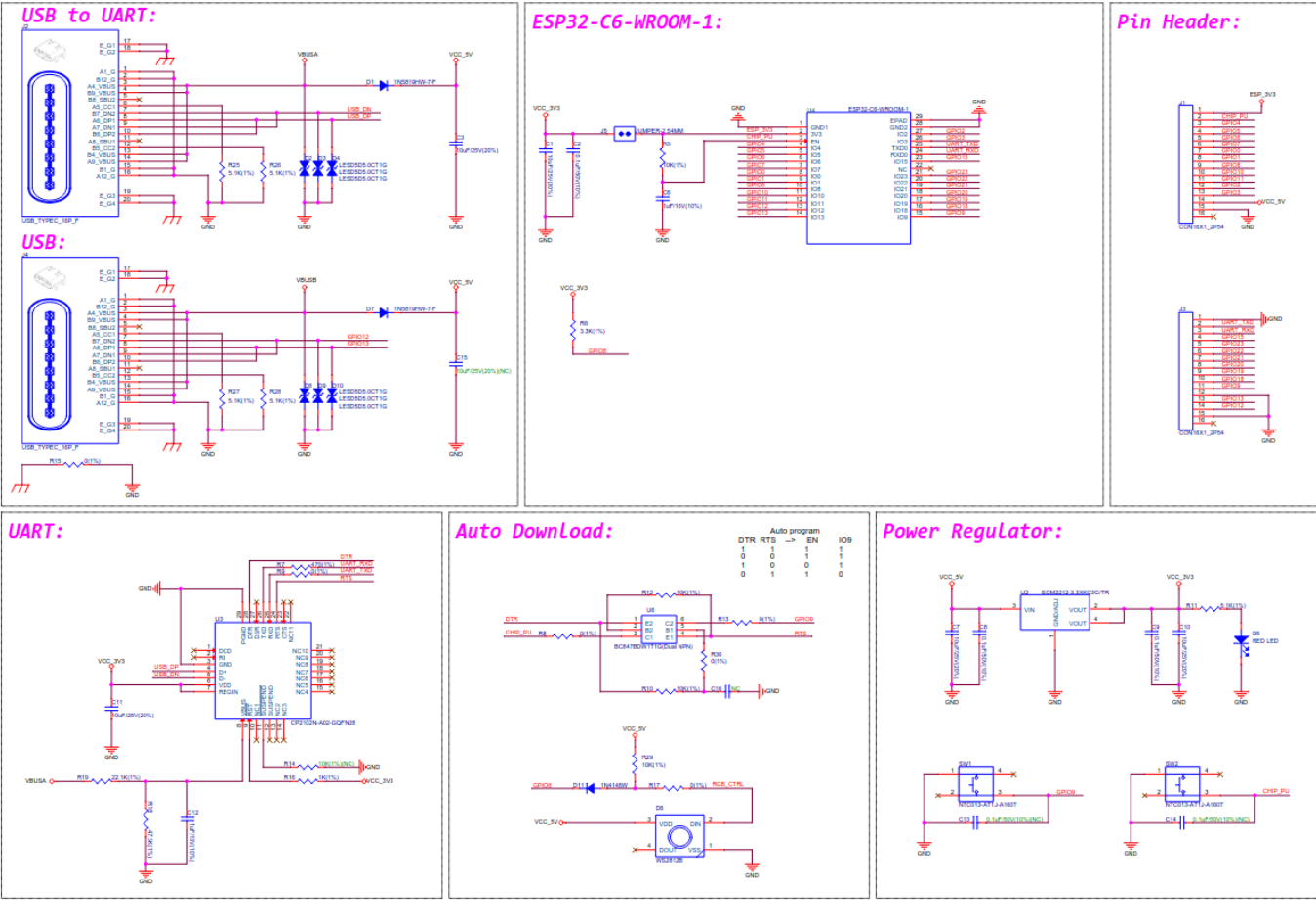


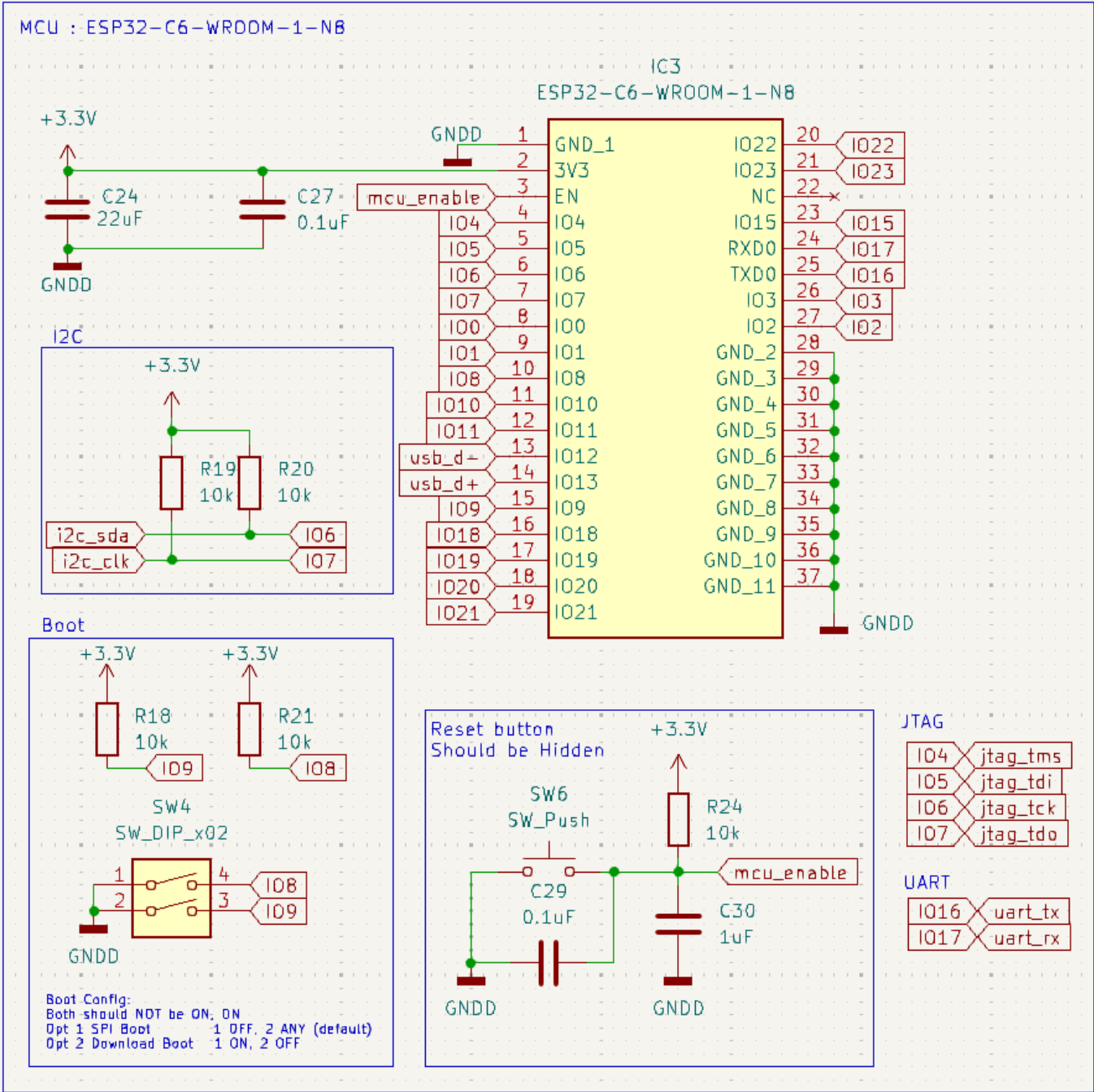
Figure 7: Peripheral Schematics

- Soldering the EPAD to the ground of the base board is not a must, however, it can optimize thermal performance. If you choose to solder it, please apply the correct amount of soldering paste. Too much soldering paste may increase the gap between the module and the baseboard. As a result, the adhesion between other pins and the baseboard may be poor.
- To ensure that the power supply to the ESP32-C6 chip is stable during power-up, it is advised to add an RC delay circuit at the EN pin. The recommended setting for the RC delay circuit is usually $R = 10\text{ k}\Omega$ and $C = 1\text{ }\mu F$. However, specific parameters should be adjusted based on the power-up timing of the module and the power-up and reset sequence timing of the chip. For ESP32-C6's power-up and reset sequence timing diagram, please refer to [ESP32-C6 Series Datasheet](#) > Section Power Supply.

A good resource is also the [ESP32-C6-DevKitC-1 v1.4](#)



Based on this two schematics, the schematic done is:



To boot the option are based on table 7 from the ESP32-C6-WROOM-1 datasheet:

Table 7: Boot Mode Control

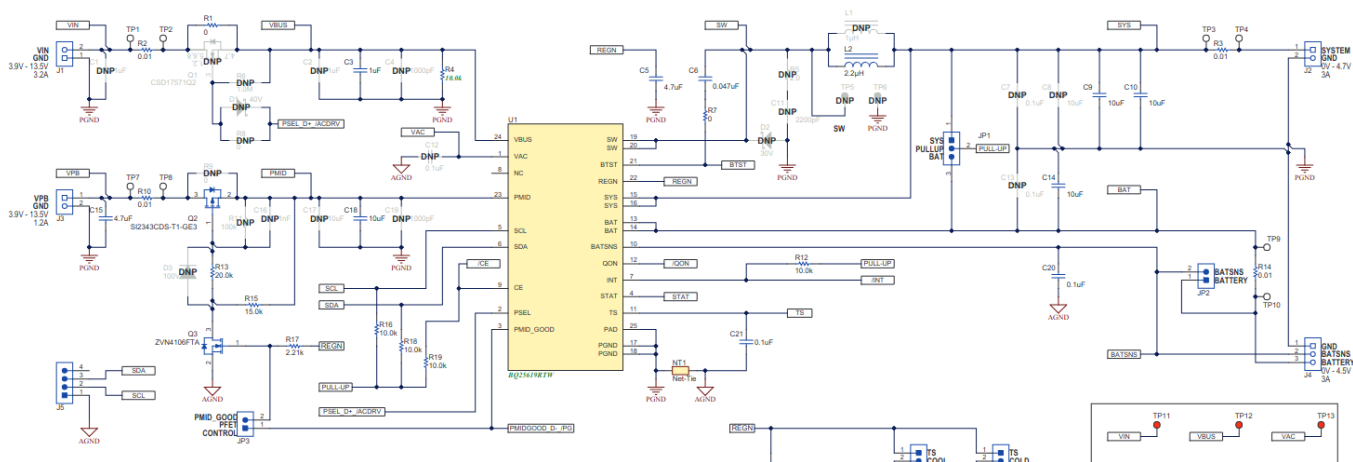
Boot Mode	GPIO8	GPIO9
Default Configuration	– (Floating)	1 (Pull-up)
SPI Boot (default)	Any value	1
Download Boot	1	0
Invalid combination ¹	0	0

¹ This combination triggers unexpected behavior and should be avoided.

4.5 PMIC (BQ25619RTWR)

The typical application from the BQ25619RTWR datasheet is:

Figure 8-1. BQ25619 Application Diagram with Optional PMOS



The schematic diagram illustrates the PMIC (BQ25619) circuit, showing various input and output connections, including Vsystem led, On/Off button, PSEL input current limit, and the main PMIC block with its pins and components.

Vsystem led: The SYSTEM pin is connected to a 10k resistor (R2) and an LED (D8) in series with the STAT pin.

On/Off button: The SW2 Push button is connected to the nQON pin through a 0.1uF capacitor (C32).

PSEL input current limit: The PSEL pin is connected to a 10k resistor (R3) and a jumper (JP2) labeled Jumper_3_Open. The other end of the jumper is connected to the PSEL pin through a 10k resistor (R27).

PMIC (BQ25619) Pin Connections:

- REGN: Connected to the REGN pin.
- +5V: Connected to the +5V pin.
- GND: Connected to the GND pin.
- C30 (4.7uF) and C31 (10uF): Connected to the GND pin.
- C26 (1uF): Connected to the GND pin.
- C28 (47nF): Connected to the BTST pin.
- L1 (1uH): Connected to the BTST pin.
- C29 (10uF): Connected to the BTST pin.
- SYSTEM: Connected to the SYSTEM pin.
- BAT+: Connected to the BAT_2 pin.
- BAT-: Connected to the BAT_1 pin.
- PGND_1: Connected to the PGND_1 pin.
- PGND_2: Connected to the PGND_2 pin.
- TS: Connected to the TS pin.
- nQON: Connected to the nQON pin.
- pmic_nINT: Connected to the nINT pin.
- pmic_nCE: Connected to the nCE pin.
- BAT+: Connected to the BAT_2 pin.
- TS: Connected to the TS pin.

4.6 LDO (TLV758P)

The typical application is:

7.1 Application Information

7.1.1 Adjustable Device Feedback Resistors

Figure 7-1 shows that the output voltage of the TLV758P can be adjusted from 0.55 V to 5.5 V by using a resistor divider network.

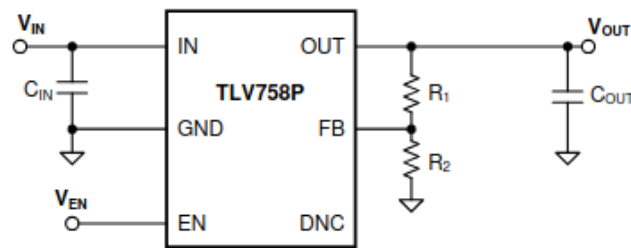


Figure 7-1. Adjustable Operation

The adjustable-version device requires external feedback divider resistors to set the output voltage. V_{OUT} is set using the feedback divider resistors, R_1 and R_2 , according to the following equation:

$$V_{OUT} = V_{FB} \times (1 + R_1 / R_2) \quad (2)$$

For this device, $V_{FB} = 0.55$ V.

To ignore the FB pin current error term in the V_{OUT} equation, set the feedback divider current to 100 times the FB pin current listed in the *Electrical Characteristics* table. This setting provides the maximum feedback divider series resistance, as shown in the following equation:

$$R_1 + R_2 \leq V_{OUT} / (I_{FB} \times 100) \quad (3)$$

For this device, $I_{FB} = 10$ nA.

7.1.2 Input and Output Capacitor Selection

The TLV758P requires an output capacitance of 0.47 μ F or larger for stability. Use X5R- and X7R-type ceramic capacitors because these capacitors have minimal variation in value and equivalent series resistance (ESR) over temperature. When choosing a capacitor for a specific application, pay attention to the dc bias characteristics for the capacitor. Higher output voltages cause a significant derating of the capacitor. For best performance, the maximum recommended output capacitance is 220 μ F.

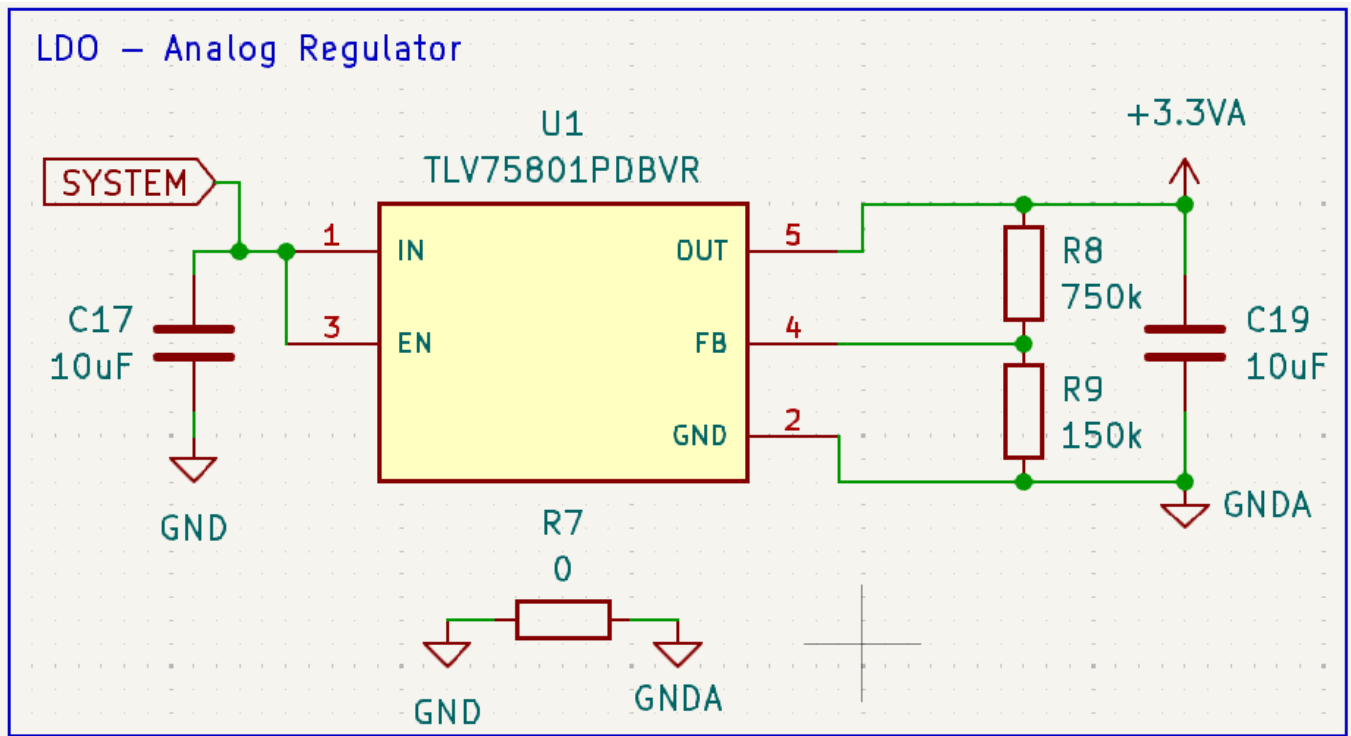
Although an input capacitor is not required for stability, good analog design practice is to connect a capacitor from IN to GND. Some input supplies have a high impedance, thus placing the input capacitor on the input supply helps reduce the input impedance. This capacitor counteracts reactive input sources and improves transient response, input ripple, and PSRR. If the input supply has a high impedance over a large range of frequencies, several input capacitors can be used in parallel to lower the impedance over frequency. Use a higher-value capacitor if large, fast, rise-time load transients are anticipated, or if the device is located several inches from the input power source.

The target analog voltage is 3.3V.

$$R1/R2 = V_{out}/V_{fb} - 1 = 5$$

$$R1 + R2 \leq V_{out}/(1e-6) = 3.3M\Omega$$

Decision was to have $R1 = 750k\Omega$ and $R2 = 150k\Omega$. Therefore, the schematic is:



4.7 Switch Regulators (TPS62A01PDDCR)

The typical application is :

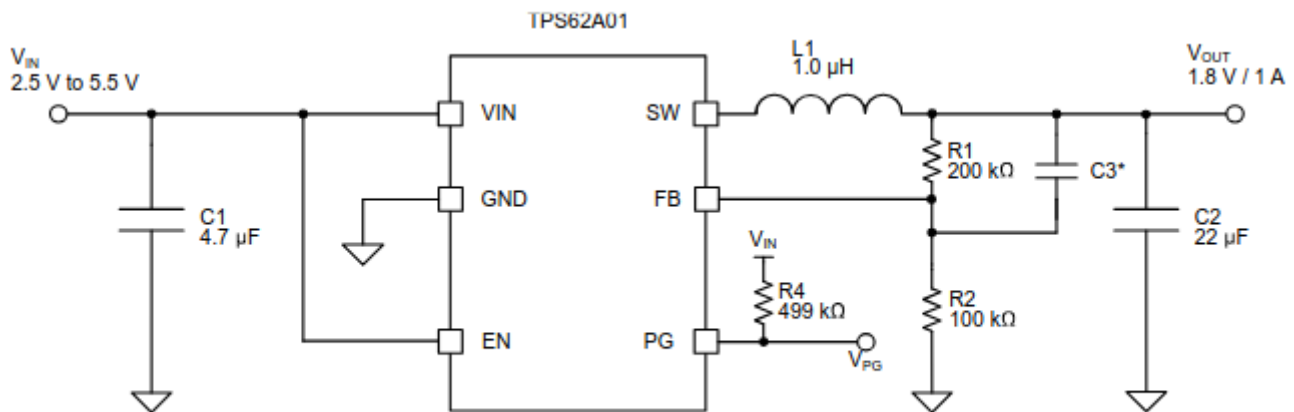


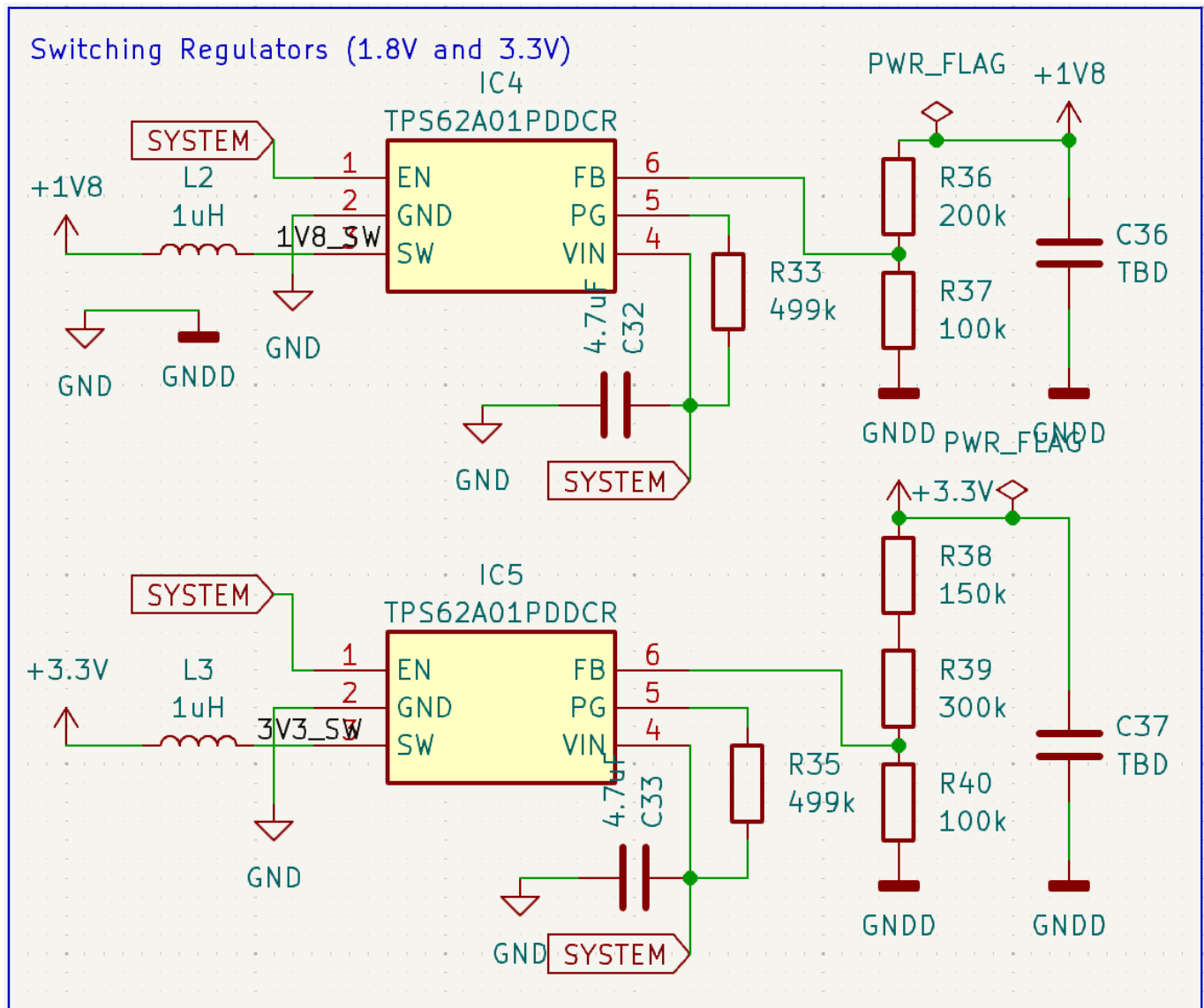
Figure 8-1. TPS62A01 Typical Application Circuit

Two regulators are required to create 1.8V and 3.3V. A voltage divider is built with $R1$ and $R2$ to fix the output voltage.

The first one can follow the same design as the typical application since it was already designed for $V_{out} = 1.8V$.

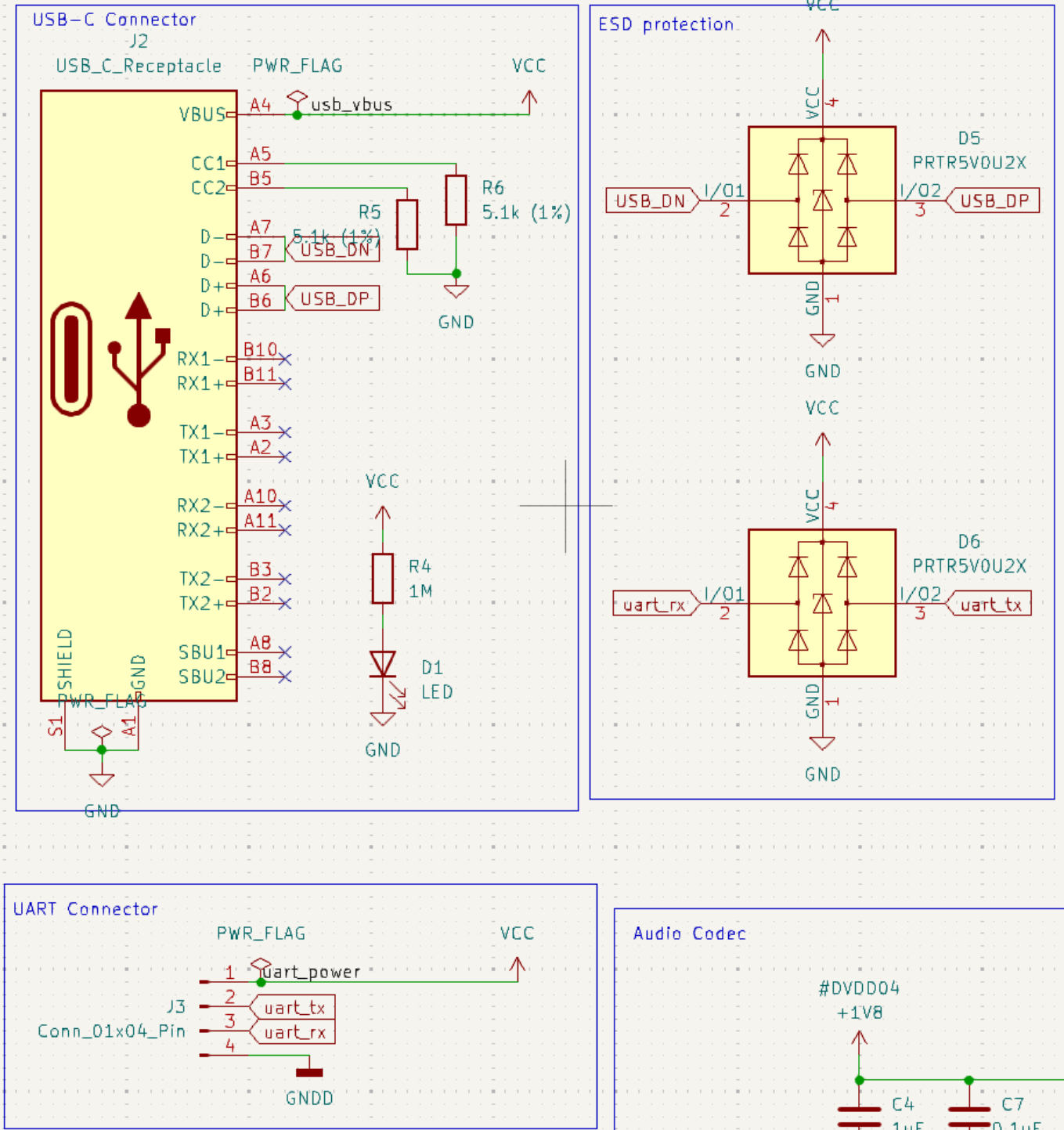
The second one needs to be designed for $V_{out} = 3.3V$. The section **8.2.2.1 Setting the Output Voltage** of the datasheet gives $R2 \leq 100k\Omega$ and $R1 = R2(V_{out}/0.6 - 1) = 4.5 \cdot R2$

Choice is made to take $R1 = 100k\Omega$, and $R2$ made with two resistors $R2 = 300k + 150k$

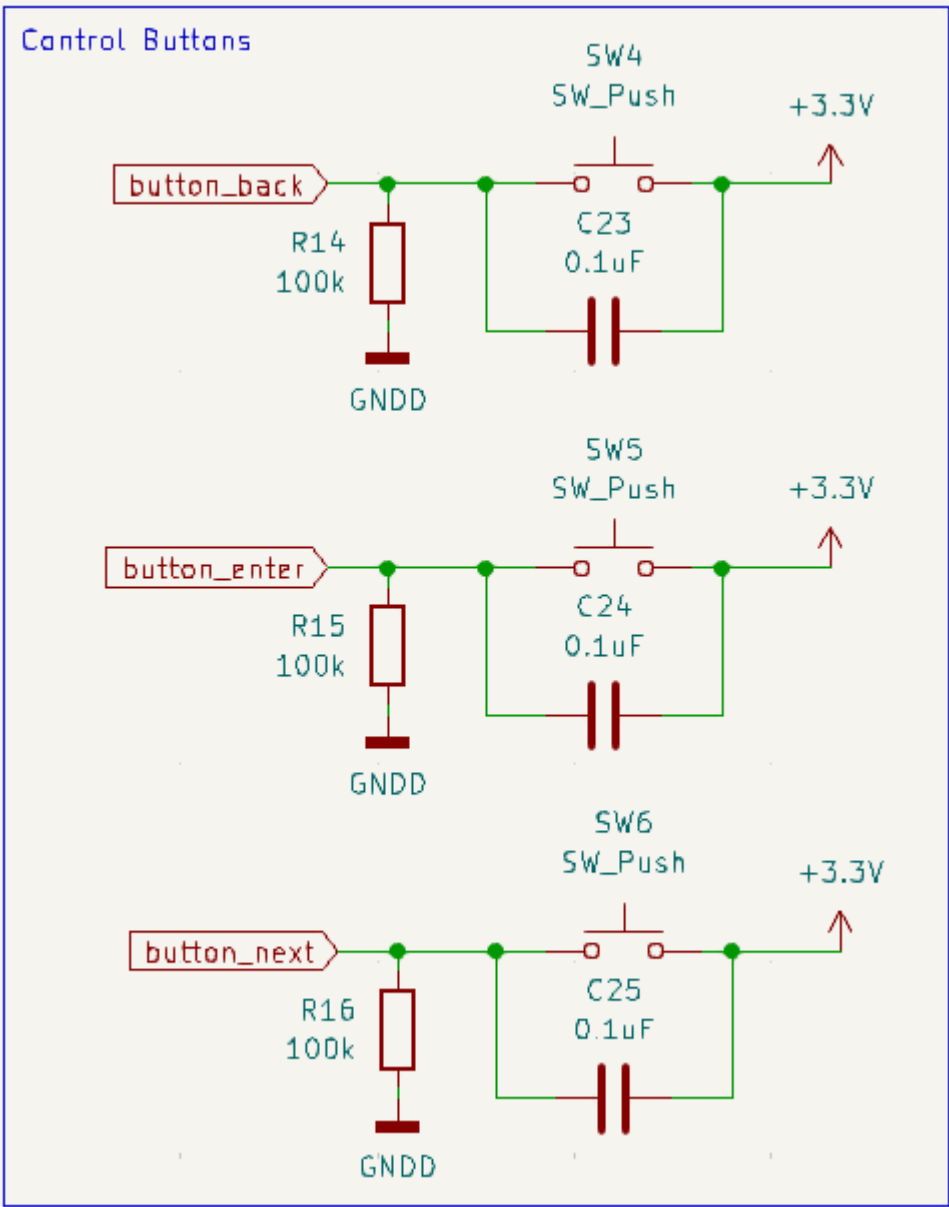


4.8 Screen

The screen is going to be connected through an SPI interface. It will be powered thanks to the 3.3 voltage. However to be able to power off the screen, an electronic circuit is required.



4.10 Control Buttons



5. GPIOs Pin Assignments

This table contains all the pin assignments. It was done updated during layout to make rooting easier.

IO°	Active Mode	Debug/Bringup
IO0	button_enter	
IO1	button_direction	
IO2	lcd_vcc_ctrl	
IO3	lcd_scl	
IO4	button_back	jtag_tms
IO5	button_next	jtag_tdi
IO6	i2c_sda	jtag_tck
IO7	i2c_clk	jtag_tdo

IO°	Active Mode	Debug/Bringup
IO8	led_dac	boot
IO9	led_adc	boot
IO10	pmic_nINT	
IO11	pmic_nCE	
IO12	usb_d-	usb_d-
IO14	usb_d+	usb_d+
IO15	lcd_res	
IO16	lcd_sda	uart_tx
IO17	lcd_dc	uart_rx
IO18	codec_reset_l	
IO19	codec_i2s_mclk	
IO20	codec_i2s_bclk	
IO21	codec_i2s_wclk	
IO22	codec_i2s_din	
IO23	codec_i2s_dout	

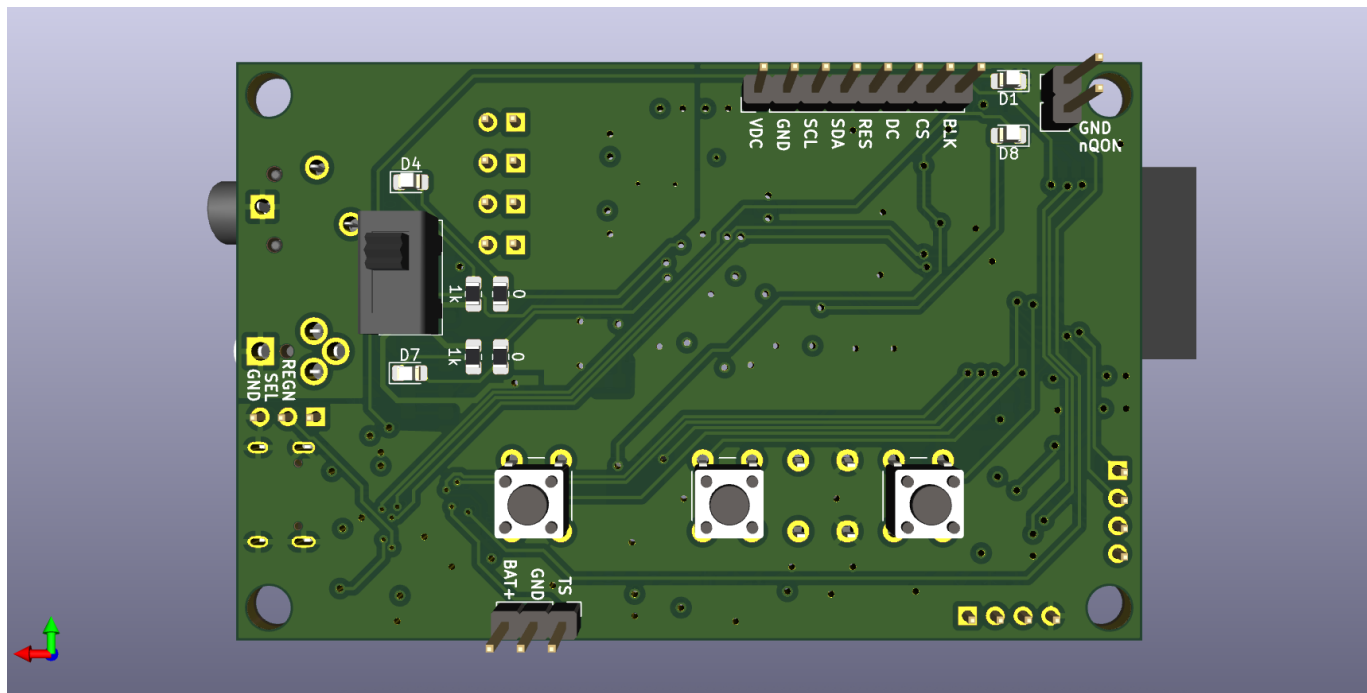
6. Layout

6.1 Layout main ideas

The goals were to have:

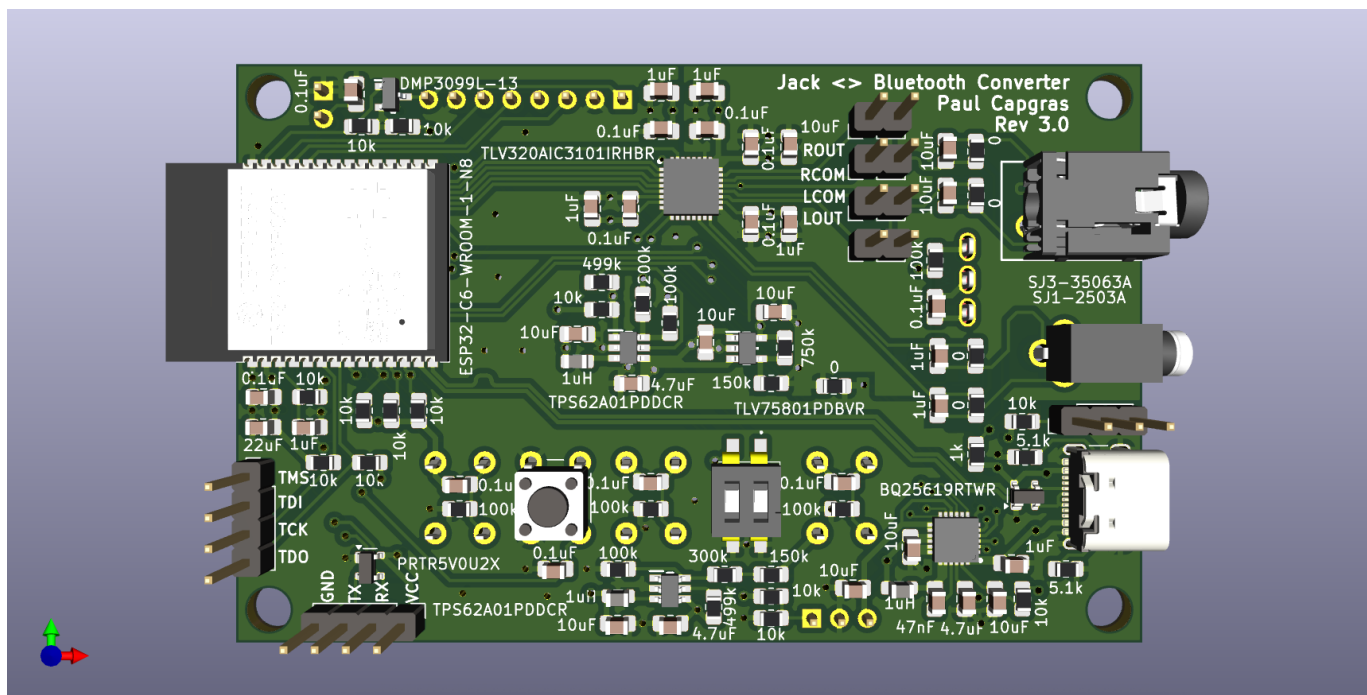
- all the high speed signals on the first layer
 - I2S
 - USB data
- all the analog signals on the first layer level and in a define region.
- respect the mecanical constraints for the buttons position. To be able to have a functional box over it.
- have the antenna on a side that could be directed toward the user
- use the led in the box.
- match impedance for the usb
- the RF should be far away from the analog
- close to the antenna, there should be no HF signals and as much ground as possible.

The user interface is the back of the PCB:



Two more leds are available on the top right corner to indicate if the device is on and the last one to indicate if the device is plugged.

On the other side, the main one, there are all the main components and all the main signals :



All the inputs/outputs connectors are located on the same slide of the board. The antenna is outside of the board to have the maximum omni-directional radiation pattern. The analog part is as far as possible from the RF.

6.2 Impedance matching

The USB differential pair should be 90ohms.

The screenshot shows the 'Calculator Tools' application window. The 'Transmission Line Type' is set to 'Coupled Microstrip Line'. The 'Substrate Parameters' are: ϵ_r : 4.5, $\tan \delta$: 0.02, p : $1.72e-08$ $\Omega \cdot m$, H : 0.2 mm, H_{t} : $1e+20$ mm, T : 0.035 mm, Roughness: 0 mm, $\mu(\text{conductor})$: 1. The 'Physical Parameters' are: W : 0.34 mm, S : 0.2 mm, L : 66.04 mm. The 'Electrical Parameters' are: Z_{even} : 60.3147 Ω , Z_{odd} : 45.1004 Ω , Ang. l: : 2.52699 rad. The 'Results' section shows: Effective ϵ_r (even): 3.57635, Effective ϵ_r (odd): 3.10677, Conductor losses (even): 0.144986 dB, Conductor losses (odd): 0.193719 dB, Dielectric losses (even): 0.210507 dB, Dielectric losses (odd): 0.184741 dB, Skin depth: 2.0873 μm , and Differential Impedance (Z_d): 90.2873 Ω . Diagrams illustrate the coupled microstrip line structure and the 'Odd' and 'Even' modes.

6.3 Trace Length

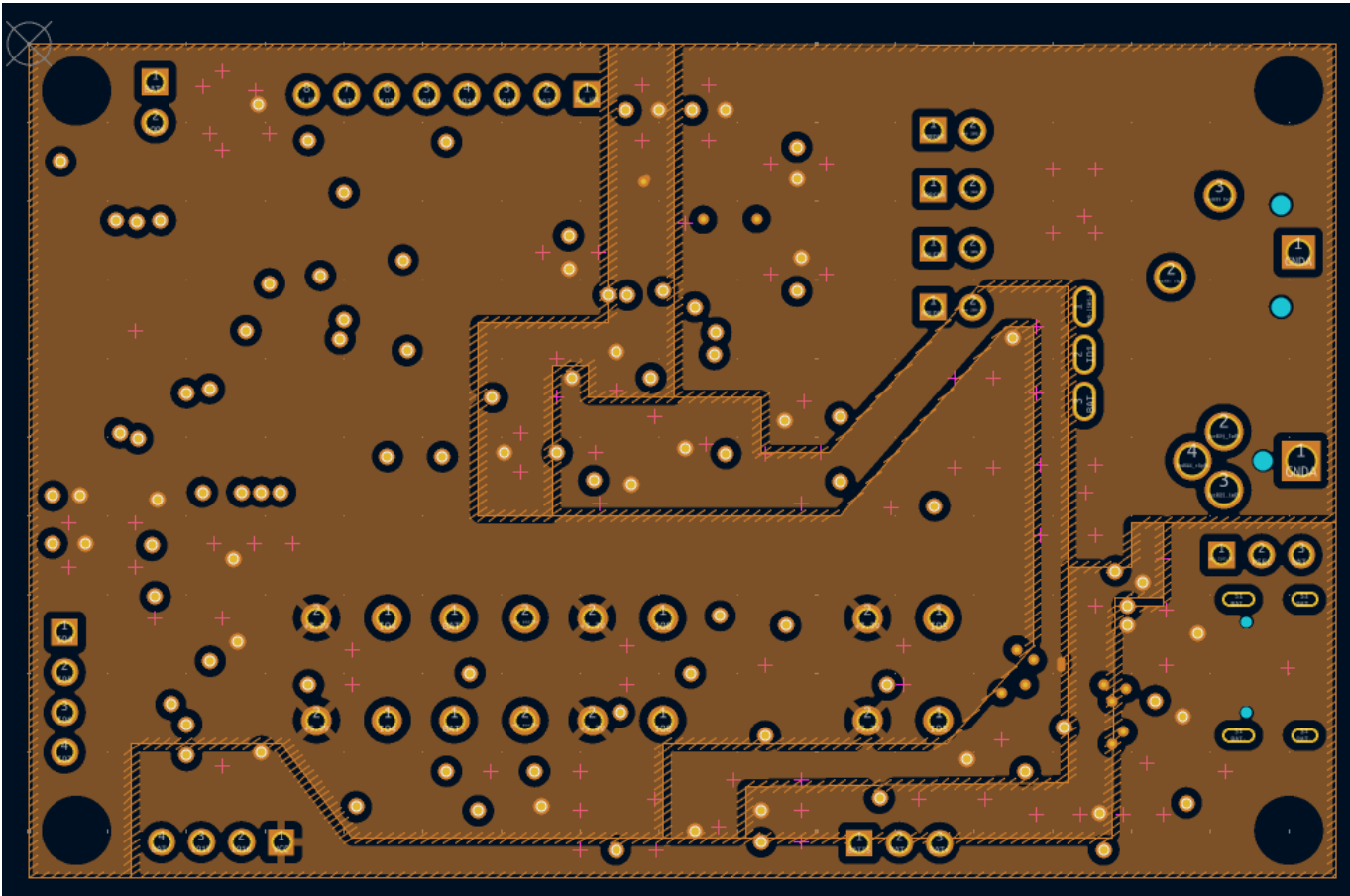
- usb are the same length (+/-0.1mm)
- left and right of jack 3.5mm are the same length (+/-0.1mm)
- left and right of jack 2.5mm are the same length (+/-0.1mm)
- for I2S:
 - Shortest line : 25.75mm
 - Longest line : 31.42mm
 - delta is 5.67mm. Assuming a delay of 6.2ps/mm, it creates a delay of 35ps which respects the timing requirements of the audio-codec.

6.4 Layers

L1 : Signals L2 : GND L3 : Power L4 : Low signals

6.4.1 Layer 1





6.4.4 Layer 4

